



Indian Institute of Technology Delhi

Cost-effective and Functional Goniometer Development

CLD 411 BTP Mid Semester Report

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Abstract

This project details the development of a cost-effective, high-accuracy goniometer software. Goniometers are crucial instruments for measuring contact angles and surface tension, but their prohibitive cost—with imported models reaching INR 10-15 lakhs and indigenous ones at INR 5-7 lakhs—creates significant barriers to accessible education and research in India. My project directly addresses this issue by employing a robust methodology based on conventional numerical methods, with a specific focus on the numerical polynomial fitting and advanced automatic baseline detection. The objective is to produce a fully functional indigenous software solution for the combined hardware for less than INR 25,000, which is more than 20 times cheaper than current market options. This initiative aligns with the "Make in India" vision and aims to democratize scientific research by providing an intuitive, efficient, and customizable tool that minimizes human intervention. The final software will be validated against commercial systems to prove its accuracy and reliability, demonstrating that high-precision scientific instrumentation can be made accessible to a wider audience.



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1 Introduction

The high cost of sophisticated scientific instrumentation remains a persistent obstacle to democratizing research and education in India. Specialized tools like goniometers, which are essential for characterizing solid-liquid interactions by measuring **contact angle** and **surface tension**, are a prime example. I have witnessed firsthand how these economic barriers stifle innovation and limit educational opportunities. A survey of the current market reveals a startling disparity: indigenous goniometers are priced at **INR 5-7 lakhs**, while imported models can cost upwards of **INR 10-15 lakhs**. This leaves many academic and research labs unable to afford the equipment necessary for fundamental studies in material science, coatings, and surface chemistry.

My project is a direct response to this challenge. My goal is to break this economic barrier by developing a fully functional, indigenous goniometer software for just **INR 25,000**, with a more advanced version for research labs priced at under **INR 40,000**. This makes my proposed system more than 20 times cheaper than the most expensive imported models. To achieve this, I will build the entire software from scratch, avoiding expensive third-party licenses and providing a solution that is both customizable and highly efficient. The software will feature a clean, intuitive GUI designed to minimize user intervention, making the analysis process more streamlined and reliable. By focusing on local development and manufacturing, this project aligns perfectly with the "Make in India" initiative, ensuring that our scientific tools are not only affordable but also contribute to the nation's technological independence.

1.1 Core Concepts and Terminology: The Language of Interfaces

Goniometry, specifically optical drop shape analysis, is a powerful technique for characterizing the surface properties of materials. At its core, the field is concerned with quantifying the interactions at the three-phase boundary where a liquid, a solid, and a vapor (or a second immiscible liquid) meet [9]. A central metric is the **contact angle** (θ), which provides a direct measure of a solid's wettability by a liquid [5, 12]. A low contact angle (less than 90°) indicates a **hydrophilic** or wettable surface, while a high contact angle (greater than 90°) signifies a **hydrophobic**, or non-wettable, surface. Extreme cases, such as the "lotus effect" observed on superhydrophobic materials, can result in contact angles approaching the theoretical limit of 180° .

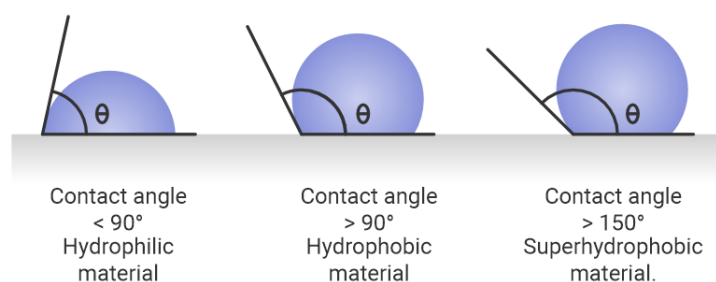


Figure 1: Illustration of a sessile drop with hydrophilic and hydrophobic contact angles (Reference: Measurlabs)

The shape of a liquid drop is a direct consequence of the interplay between various forces, primarily **surface tension** and **gravity** [4]. Surface tension (γ_{LV}) is a manifestation of the cohesive forces between liquid molecules. In the context of a solid surface, this gives rise to additional interfacial tensions: the solid-vapor interfacial tension (γ_{SV}) and the solid-liquid interfacial tension (γ_{SL}). Optical goniometers, also known as drop shape analyzers, capture images of drops and analyze their profiles to calculate these parameters. The primary optical measurement techniques include the **sessile drop method** for static contact angles and surface energy determination, and the **pendant drop method** for measuring surface or interfacial tension. For a more complete characterization of real-world surfaces, **dynamic methods** are employed to measure advancing and receding contact angles as the drop volume is increased or decreased, revealing the crucial **contact angle hysteresis**.

1.2 The Physics of Drop Shape: From Young to Young-Laplace

The theoretical foundation of contact angle measurement is **Young's equation**, which describes the mechanical equilibrium of a drop on an ideal solid surface [2].

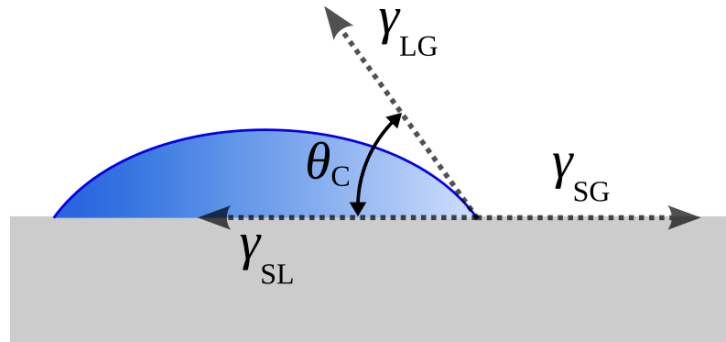


Figure 2: Schematic of a liquid drop showing the quantities in the Young equation.
(Reference: Wikipedia)

The equation, expressed as

$$\gamma_{SV} = \gamma_{SL} + \gamma_{LV} \cos \theta_Y$$

represents a balance of forces at the three-phase contact line, leading to the theoretical **equilibrium contact angle** (θ_Y). This model, however, rests on a series of idealized assumptions: that the solid surface is perfectly smooth, rigid, and chemically homogeneous [2].

For drops with significant volume, gravity plays a crucial role, deforming the drop from a simple spherical cap. To account for this, goniometry software employs a more sophisticated model based on the **Young-Laplace equation** [11]. This equation provides a relationship between the pressure difference across a curved fluid interface, the surface tension, and the principal radii of curvature (r_1 and r_2). For a sessile drop, the hydrostatic pressure due to the drop's weight varies with height, causing the surface curvature to change along the drop profile. The **Young-Laplace fit** is a numerical method that determines the parameters of an equation system modeling the drop shape, allowing for

accurate contact angle and surface tension measurements by finding the intersection of the modeled contour with the baseline [11]. The use of this physics-based model is essential for achieving the highest possible accuracy, particularly for larger drops and large contact angles where the shape deviates significantly from a simple arc or ellipse.

1.3 The Inherent Challenges of Real-World Measurement

A key challenge in developing robust goniometer software is that the idealized assumptions of Young's equation are almost never met in practice [2]. The surfaces of real materials are often chemically and topographically heterogeneous. This means that a single static contact angle measurement may not provide a complete picture of the surface's behavior, leading to the need for dynamic measurements that reveal contact angle hysteresis [2]. This inherent complexity of real-world samples is the primary reason why sophisticated, multi-faceted algorithms are not just a luxury, but a necessity. The pursuit of "robust and accurate" software must begin by accepting that the foundational theory is an ideal, and the algorithms must be designed to compensate for the reality of imperfect data and non-ideal drop shapes. This need to bridge the gap between a simple theoretical model and a complex practical reality drives the design of all advanced goniometry software, from commercial systems to open-source solutions.

2 Literature Review: The Software Ecosystem

2.1 Conventional Numerical Algorithms: The Cornerstone of Accuracy

Traditional goniometer software relies on a pipeline of image processing and numerical fitting algorithms to determine contact angles and surface tension. This approach, while well-established, presents a fundamental trade-off between algorithmic flexibility and the accuracy of the underlying physical model.

2.1.1 Geometric and Tangent Fitting Methods

These methods assume that the drop profile can be approximated by a simple geometric shape, and the contact angle is calculated from the tangent at the intersection of this shape and the baseline.

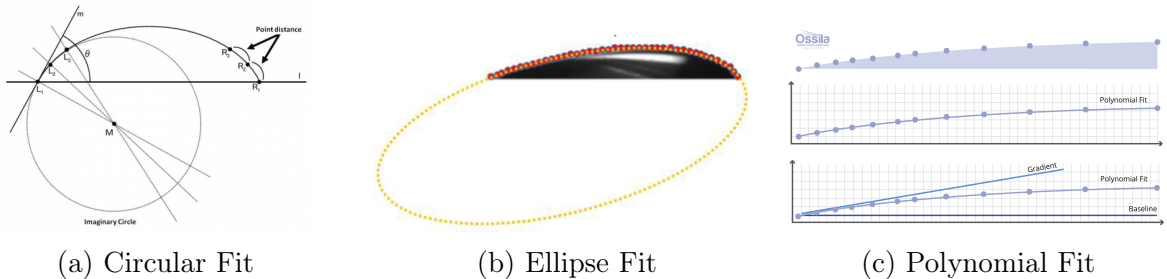


Figure 3: Comparison of fitting methods: (a) Circular, (b) Ellipse, and (c) Polynomial. (Reference: (a) Kyowa (b) Research Gate (c) Ossila)



- **The Circle and Conic Section Methods:** The simplest approach is to fit a circular arc or an ellipse to the drop profile [10]. The circle method is only suitable for small drops (0° to 20°) where gravity is negligible. The conic section method, which assumes an elliptical arc, is more accurate and has a wider application range (10° to 100°), as it better accounts for the slight flattening of the drop due to its own weight.
- **The Polynomial Fitting Method:** A more flexible approach, it works by first extracting the drop contour and then fitting a polynomial function to the profile data points. The contact angle is then calculated from the slope of the polynomial's first derivative at the contact point. This method can handle drops that are not perfectly symmetrical. A common polynomial form is:

$$y(x) = ax^n + bx^{n-1} + \dots + c$$

The significant drawback is that the accuracy is highly sensitive to the order of the polynomial (n) and the number of pixels used for the fitting procedure. An incorrect choice can lead to inaccurate extrapolation of the drop shape, compromising the measurement.

2.1.2 Physics-Based Analysis: The Young-Laplace Gold Standard

The most accurate and widely accepted conventional method for drop shape analysis is the **Axisymmetric Drop Shape Analysis (ADSA)** algorithm. ADSA is a physics-based approach that numerically integrates the Young-Laplace equation to generate a theoretical drop profile. The equation is expressed as:

$$\Delta P = \gamma \left(\frac{1}{R_1} + \frac{1}{R_2} \right)$$

The algorithm then uses an iterative process to find the set of physical parameters (e.g., surface tension, γ) that produce a theoretical curve that best matches the experimentally observed drop contour [2]. This method is considered the gold standard for its precision. Its primary strength is its unparalleled accuracy when a symmetrical, undistorted drop is present on a level, smooth surface. The method is inherently fragile and not suitable for dynamic or asymmetrical drops [5].

2.2 The Deep Learning Paradigm: A New Frontier

The advent of deep learning has introduced a new paradigm for goniometry, moving beyond the traditional pipeline of image processing and numerical fitting. Deep learning models, particularly Convolutional Neural Networks (CNNs), are not just an alternative; they represent a fundamentally different approach that addresses some of the most persistent challenges in the field, such as speed and robustness to non-ideal conditions.

2.2.1 Beyond Fitting: Direct Prediction from Raw Images

Deep learning models can be trained to directly predict a physical property, such as contact angle or surface tension, from a raw image without the need for manual intervention or a multi-step analytical process. This approach bypasses the entire conventional

pipeline, which typically involves edge detection, baseline fitting, and then fitting a numerical model to the profile. Research has demonstrated that CNNs can accurately predict surface tension values from raw pendant drop images, even those captured with a common smartphone, with high accuracy. The primary advantage of this methodology is its exceptional speed and robustness to image imperfections like suboptimal lighting, glare, or optical aberrations.

2.2.2 Synthetic Data Generation with GANs

Deep learning models are notoriously data-hungry, requiring vast, high-quality labeled datasets to achieve optimal performance. This presents a significant bottleneck for applications in specialized fields like goniometry. **Generative Adversarial Networks (GANs)** offer a powerful solution to this challenge.

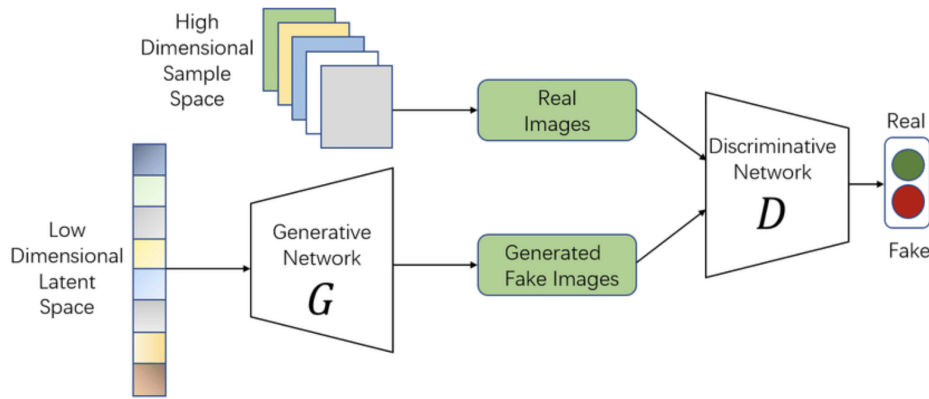


Figure 4: The architecture of a Generative Adversarial Network (GAN) framework.
(Reference: LABELERR)

A GAN consists of two competing neural networks: a **generator** and a **discriminator**. The generator creates synthetic data from a random input, while the discriminator's role is to evaluate whether a given data sample is real or fake. This adversarial process drives both networks to improve until the generator can produce synthetic data that is indistinguishable from the real data. For goniometry, training a deep learning model on a combination of a small real dataset and a large synthetic dataset generated by a GAN can dramatically improve the model's performance and generalization without the prohibitive cost of collecting millions of real images.

2.3 The Critical Challenge of Automatic Baseline Detection

Before any contact angle or surface tension calculation can be performed, goniometer software must accurately identify the **baseline**, or the solid-liquid interface. The failure to do so is a single point of failure that will render any subsequent algorithmic analysis useless. A robust solution must, therefore, prioritize the reliability of this crucial first step.

2.3.1 Conventional Computer Vision Techniques

Conventional approaches to automatic baseline detection typically rely on identifying sharp changes in pixel intensity that correspond to the drop-solid interface.

- **Gradient-Based Edge Detection:** The baseline is a horizontal line, which corresponds to a vertical edge in a profile image. This edge can be identified by detecting a high vertical gradient in pixel intensity. Algorithms such as **Canny edge detection** are commonly used for this purpose.
- **Robust Linear Regression (Theil-Sen Estimator):** After edge detection, a set of pixels representing the potential baseline is obtained. The **Theil-Sen estimator** is a non-parametric statistical method for robustly fitting a line by selecting the median of the slopes of all lines through pairs of points. This approach is significantly more accurate than least squares for noisy and heteroskedastic data.
- **Parametric Line Detection (Hough Transform):** The **Hough Transform** is an alternative to regression for finding geometric shapes like lines and circles from a collection of edge points. It works by transforming each edge point from image space into a parameter space, where points corresponding to a straight line accumulate "votes" in a single cell. A major advantage of this method is its tolerance for gaps and noise in the feature boundary.

2.3.2 Semantic Segmentation with Deep Learning

A more advanced approach, enabled by deep learning, is **semantic segmentation**. Semantic segmentation is a computer vision task that classifies every pixel in an image into a predefined category, such as "droplet," "baseline," or "background".

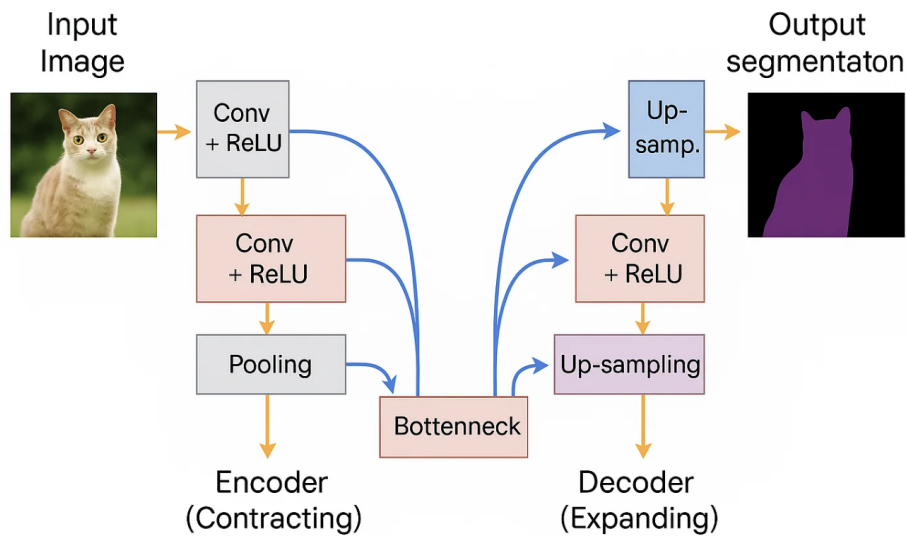


Figure 5: The U-Net architecture for semantic segmentation of front from background.
(Reference: Great Learning)

The **U-Net** architecture, with its symmetric encoder-decoder structure and "skip connections," is particularly effective for this task. The model develops a semantic un-



derstanding of what a "baseline" is, allowing it to accurately detect the solid-liquid interface even in the presence of complex backgrounds, reflections, or poor lighting. The pixel-perfect mask generated by the model can then be used to extract a clean profile for downstream analysis, whether it be a conventional fitting method or a direct deep learning prediction.

A key limitation of the U-Net architecture, however, is its **reliance on large, meticulously labeled datasets** for training. Creating a pixel-by-pixel mask for thousands of images is a labor-intensive process that can be a significant bottleneck for specialized applications where data is scarce. Furthermore, the model's ability to generalize is highly dependent on the diversity of its training data; it may fail to perform accurately on images captured under conditions not represented in its training set.

2.3.3 Conclusion

In conclusion, the literature reveals that goniometry software is a complex ecosystem defined by a fundamental tension between **algorithmic accuracy**, **robustness to real-world conditions**, and **economic accessibility**. While licensed commercial solutions offer integrated workflows and high-precision, they are prohibitively expensive. Open-source alternatives, though free, often lack the polish and automation needed for efficient research. No single conventional algorithm is perfect; methods like ADSA offer unparalleled accuracy for ideal drops, while polynomial fitting and deep learning excel at handling the noise and imperfections of real-world data. Therefore, a successful, affordable, and indigenous solution must synthesize these strengths, combining the physics-based rigor of conventional analysis with the robust automation of modern computer vision techniques, all within a clean and user-friendly interface. This hybrid approach is the only way to bridge the gap between theoretical ideals and the practical demands of democratized scientific research.



3 Methodology

My project will follow a systematic, multi-phase methodology to develop a robust and accurate goniometer software. The core of my approach is to combine the precision of physics-based numerical analysis with user-driven feedback to overcome the limitations of a limited dataset.

3.1 Software Architecture and Technology Selection

The software is being developed in **Python**, chosen for its rich ecosystem of scientific libraries and versatility. My core technology stack includes:

- **OpenCV** for all image processing tasks.
- **NumPy** for efficient numerical operations.
- **SciPy** for curve-fitting algorithms.
- **Matplotlib** for visualization and debugging.
- A simple graphical user interface (GUI) to allow user interaction and real-time visual feedback.

3.2 Algorithmic Pipeline for Contact Angle Calculation

The pipeline for calculating the contact angle is designed to be highly reliable, with user-driven steps where automation is still under development.

3.2.1 Image Acquisition and Pre-processing

The process begins with image acquisition, followed by the user selecting a **Region of Interest (ROI)** to focus the analysis on the droplet. The current version requires the user to manually select the baseline, a crucial step that ensures measurement accuracy. Following this, the image undergoes standard pre-processing: conversion to grayscale and application of a **Gaussian blur** to reduce noise.

3.2.2 Edge Detection and User-Driven Thresholding

To precisely extract the droplet profile, I am implementing a unique functionality that allows the user to interactively adjust the thresholds for **Canny edge detection**. The user can see the points detected for any given thresholds in real-time, which allows them to select the final parameters that capture the droplet's edge with the highest precision. This approach mitigates the risk of inaccurate edge detection and ensures the highest quality input for the subsequent fitting process.

3.2.3 Profile Fitting and Final Calculation

My current implementation utilizes a **second-order polynomial fitting algorithm** to model the droplet's shape. This method is computationally inexpensive and provides a good approximation for a wide range of contact angles. My final goal is to enhance this



by implementing a feature that allows the user to select from multiple fitting algorithms (e.g., polynomial, conic, etc.). The software will then automatically select the best fit based on a rigorous statistical analysis (**R² value**) to provide the most accurate result. The contact angle is then calculated from the derivative of the fitted polynomial at the three-phase contact point.

3.3 Algorithmic Pipeline for Surface Tension Calculation

The pipeline for surface tension calculation is tailored to the pendant drop method, focusing on accurate measurement of the droplet's dimensions.

3.3.1 Image Pre-processing and Profile Extraction

The process begins with image acquisition and ROI selection, followed by conversion to grayscale and application of a Gaussian blur. For edge detection, I am again using a user-driven approach with Canny edge thresholds to precisely extract the pendant drop's profile.

3.3.2 Dimensional Analysis and Surface Tension Calculation

Instead of relying on a complex Young-Laplace fitting algorithm, I am implementing a more computationally efficient framework based on the ratio of the drop's dimensions. Surface tension (γ) is estimated using the following equation:

$$\gamma = \frac{\Delta\rho g D_E^2}{H}$$

where $\Delta\rho$ is the difference in density between the liquid and the surrounding air, g is the gravitational constant, D_E is the maximum diameter of the drop, and $1/H$ is a dimensionless function of the ratio of the drop's maximum diameter (D_E) and its diameter at a distance D_E from the apex (D_S). The value for the dimensionless function $1/H$ will be numerically calculated using the following simple analytical formula:

$$\frac{1}{H} = a \left(\frac{D_S}{D_E} \right)^b$$

where $a \approx 0.345$ and $b \approx -2.5$ are constants derived from established literature [6, 4]. The software will automatically measure D_E and D_S from the extracted droplet profile to perform the calculation. This provides a robust and reliable method that does not require the computationally intensive solution of differential equations, making it ideal for a low-cost, resource-efficient system.

3.4 Future Work: Towards Full Automation

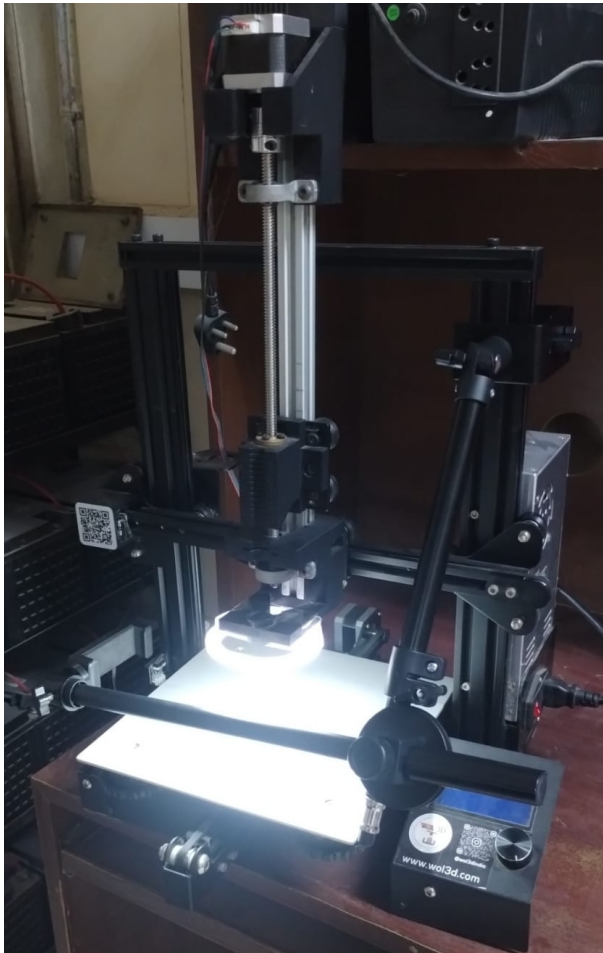
A key area for future development is the integration of an automatic baseline detection algorithm. I am currently experimenting with the **Hough Transform** to identify the baseline and also to assist in the removal of inner points by creating a binary mask to segment the drop from the background. Once this feature is robustly implemented, it will be integrated into the main pipeline to further minimize user intervention and enhance overall efficiency.

4 Description of Experimental Setup and Software Procedure

This section details the hardware and software components used in my project, along with the step-by-step procedure for goniometric measurements. My entire workflow is designed to be intuitive and efficient, ensuring reliable data acquisition and analysis.

4.1 Experimental Setup

The first prototype of my hardware was built as a proof of concept. It consisted of a smart-phone camera, a novel automated 3-axis motion stage, a custom automated microliter-precision dispensing system, blunt syringes, and an LED panel. This system, while functional, had several limitations: the phone camera's positioning significantly impacted image quality, its resolution was quite low, and the lighting was not diffused, which affected image quality a lot. The setup also required an external laptop for operation.



(a) The first hardware Prototype



(b) The camera setup for new hardware

Figure 6: Hardware setup for Image Acquisition

A new prototype design is under progress to overcome these issues. The new system integrates a **Raspberry Pi Camera Module (IMX477)** and an **Arduino UNO**. This

setup significantly improves the camera resolution and eliminates the need for a separate laptop, as a web-based GUI can be used for control. A simple light diffusing system has also been added to improve image quality. A current limitation is the manual focus of the IMX477 camera, which I plan to resolve in the future either by using a motorized hardware for user-controlled focus or by upgrading to a new Raspberry Pi camera with built-in variable focus.

4.2 Software Procedure

My software is structured into two distinct workflows, one for contact angle determination and another for surface tension calculation, to provide specialized and accurate analysis. The entire process is best visualized as a pictorial flowchart, as shown in Figure 7.

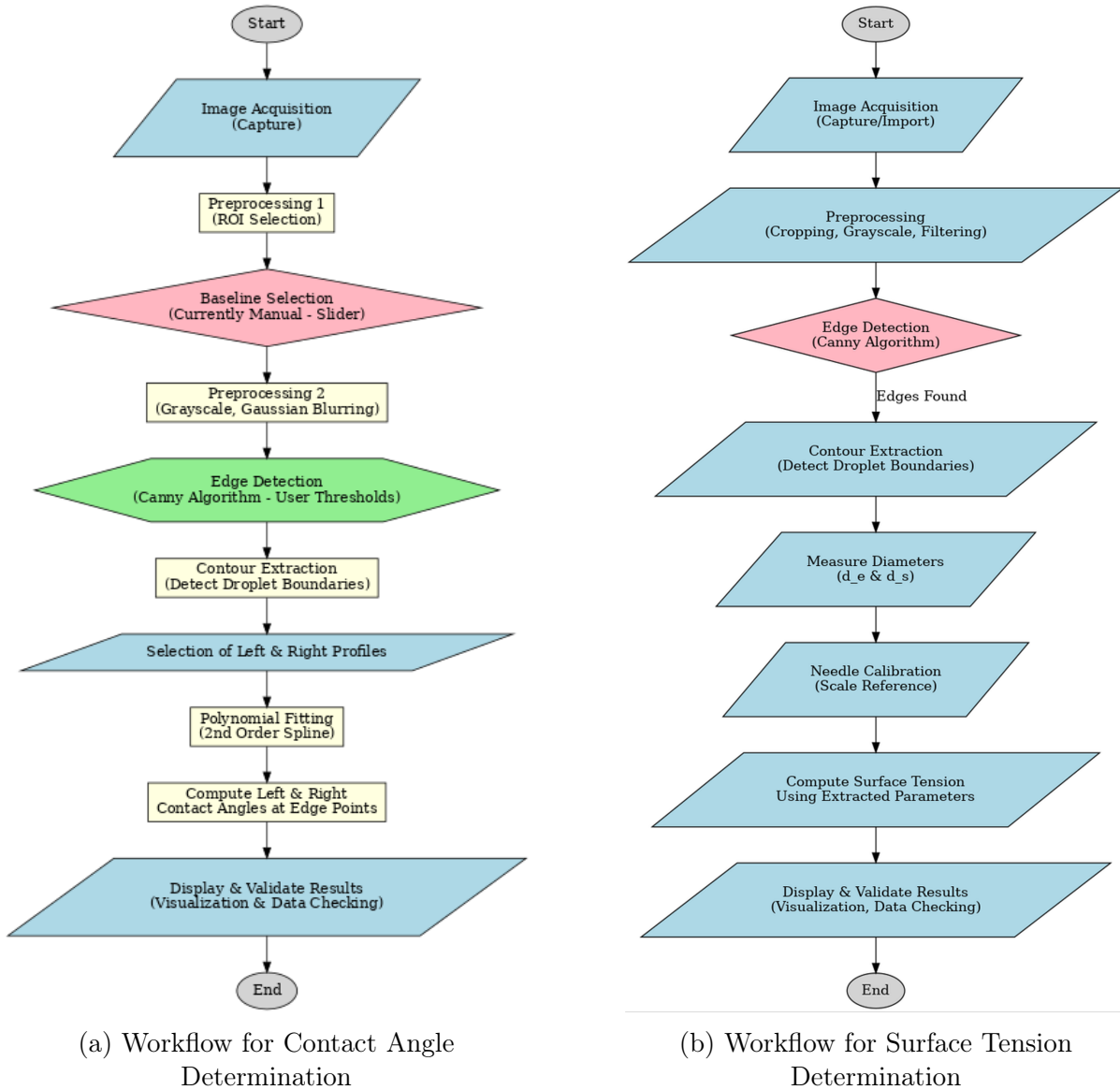
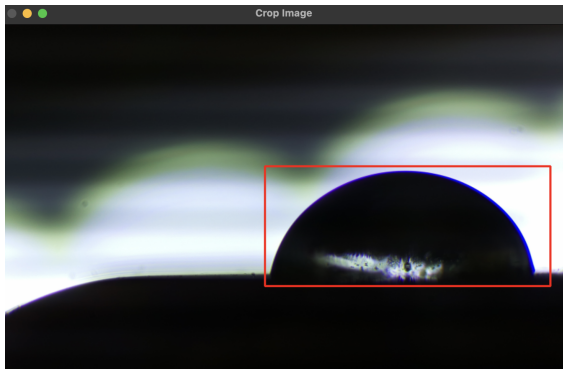


Figure 7: A comprehensive flowchart of the goniometer software workflow, detailing the steps for both contact angle and surface tension analysis.

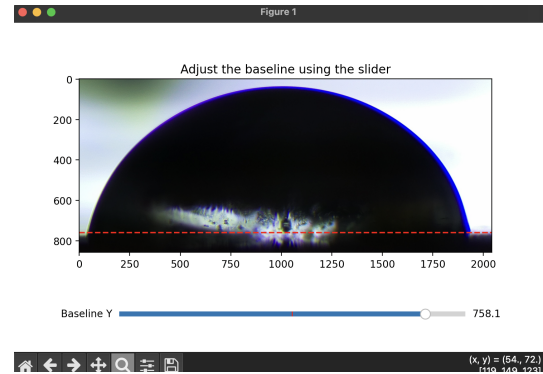
4.2.1 Workflow for Contact Angle Determination

The procedure begins with the user loading a droplet image and manually selecting the **Region of Interest (ROI)**. The image is then automatically converted to grayscale and a **Gaussian blur** is applied to reduce noise. The user then takes control, interactively adjusting the thresholds for **Canny edge detection** with real-time feedback to precisely capture the droplet's profile.

After edge detection, the user manually defines the baseline, though I am developing an automated solution using the **Hough Transform** to enhance this step. The software then uses a **second-order polynomial fitting algorithm** to model the droplet's shape and calculates the contact angle from the tangent of the fitted curve.



(a) User-enabled ROI Selection



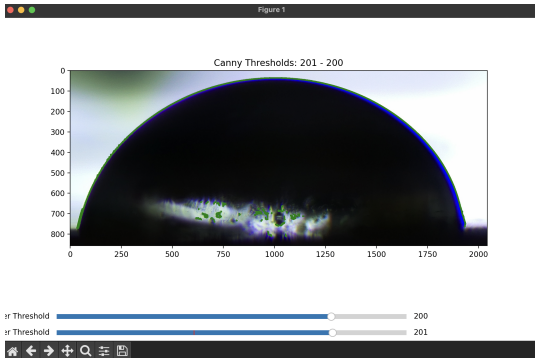
(b) Baseline Detection



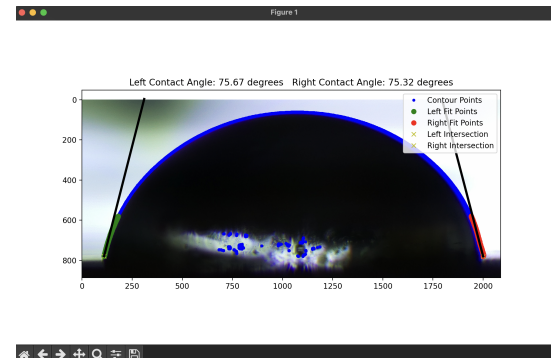
(c) Conversion to Grayscale



(d) Gaussian Blurring to filter noise



(e) Interactive Canny Edge Detection



(f) Polynomial Fitting and Final Result

Figure 8: Step-by-step visualization of the pendant drop method to compute contact angle.

4.2.2 Workflow for Surface Tension Determination

For surface tension, the process is similar but tailored for a pendant drop. The user first selects the ROI on a pendant drop image. The software handles pre-processing (grayscale and Gaussian blur) and then allows the user to interactively set the Canny edge thresholds.

Instead of a computationally intensive Young-Laplace fit, my software uses a more efficient dimensional analysis method. It automatically measures the droplet's maximum diameter (D_E) and its diameter at a distance D_E from the bottom (D_S) from the extracted profile. These values are then used in a simple analytical equation to determine the surface tension.

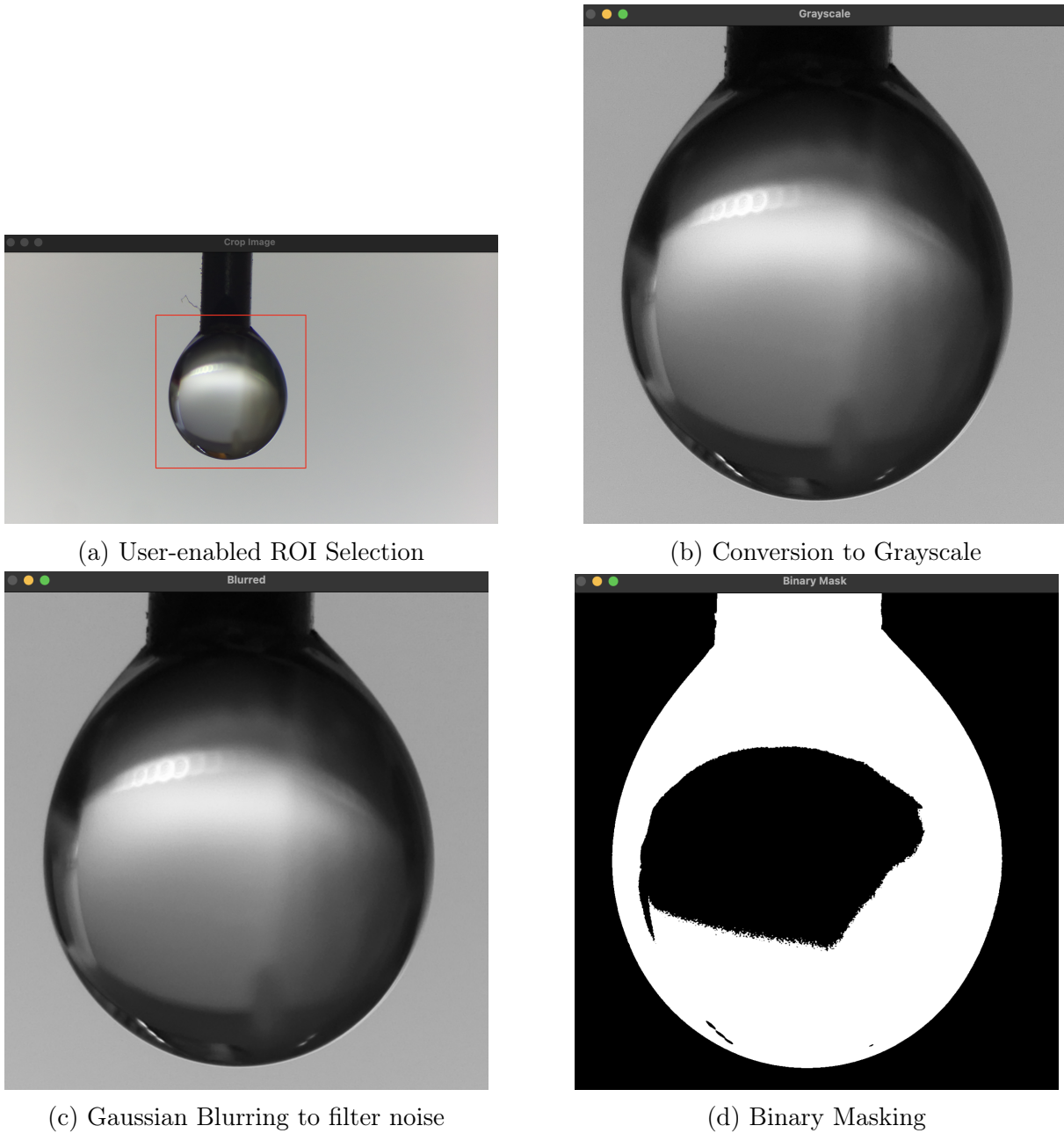
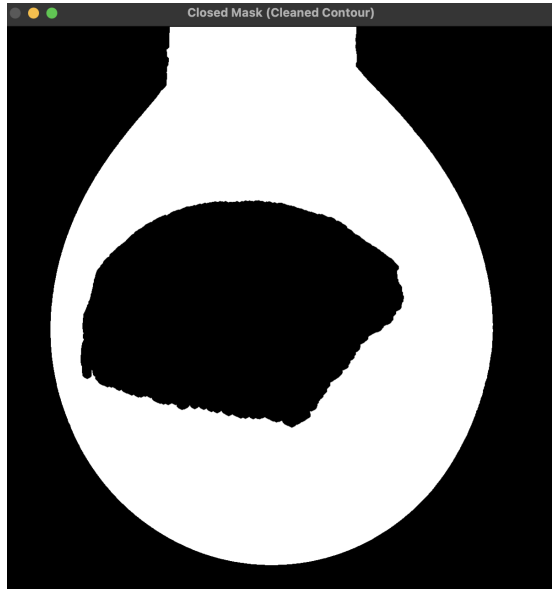
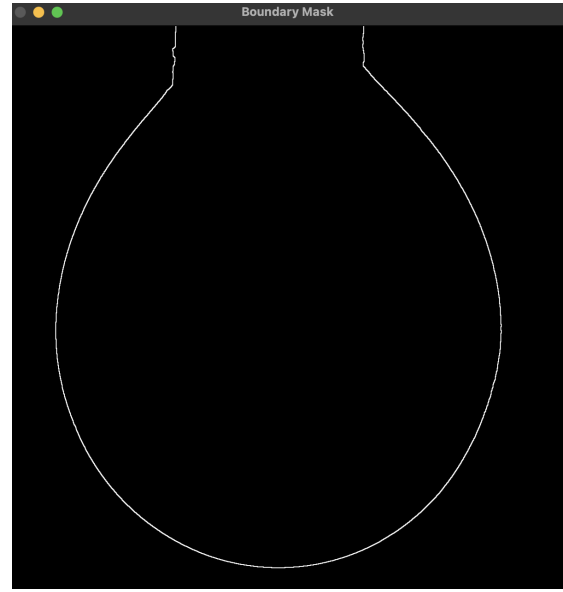


Figure 9: Step-by-step visualization of the pendant drop method to compute surface tension.



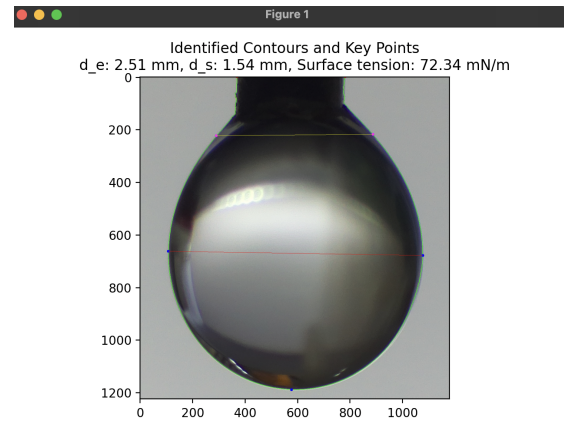
(a) Closed Morphological Masking



(b) Inner Point Removal



(c) Final Contour Detection



(d) Scaling and Final Result

Figure 10: Step-by-step visualization of the pendant drop method to compute surface tension.

Through this two-part workflow, my software demonstrates a practical and effective solution to the challenges of goniometry. By prioritizing a hybrid approach—combining low-cost, open-source hardware with precise numerical algorithms and a user-driven interface—I am directly addressing the economic barriers that limit scientific research. This detailed procedure proves that high-quality, reliable scientific instrumentation is not a luxury, but an achievable goal for local, indigenous development.

5 Results and Discussion

This section presents the key outcomes of my project's development and implementation, demonstrating how the software and hardware prototypes meet the objectives outlined in this proposal. The discussion will focus on the performance of the core algorithms and the overall viability of an indigenous, low-cost goniometer system.

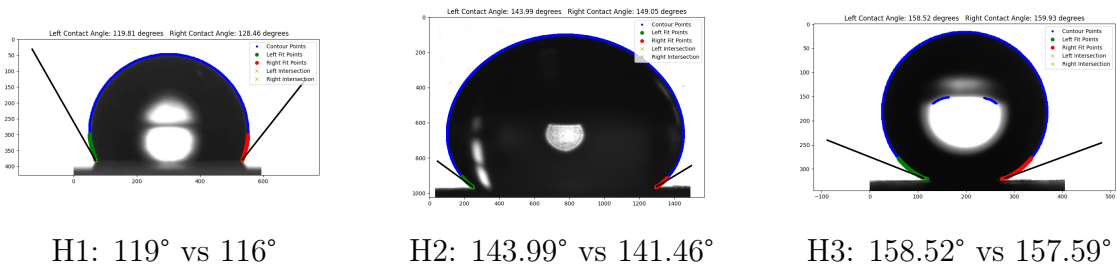
5.1 Contact Angle Determination

Ten droplets with known reference contact angles were analysed to evaluate the accuracy of the developed optical goniometer. The dataset spans three wettability regimes: (i) highly hydrophobic ($\theta > 110^\circ$), (ii) intermediate ($40^\circ < \theta < 95^\circ$), and (iii) highly hydrophilic ($\theta < 25^\circ$). This categorisation ensures testing across diverse Bond-number conditions and drop geometries.

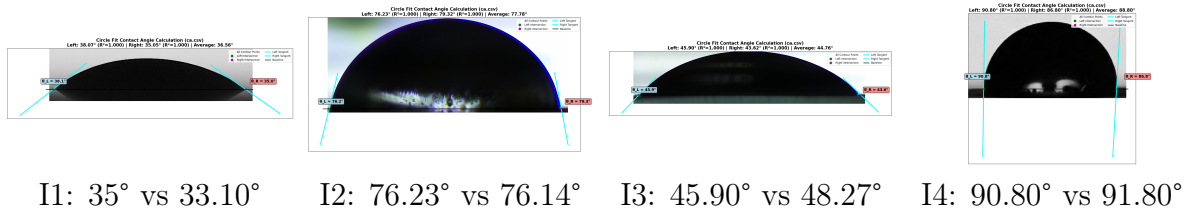
Extracted Contact Angle Outputs (All Samples)

Figure 11: Extracted droplet edges and fitted contact angles for all ten samples.

Hydrophobic ($\theta > 110^\circ$)



Intermediate ($40^\circ < \theta < 95^\circ$)



Hydrophilic ($\theta < 25^\circ$)

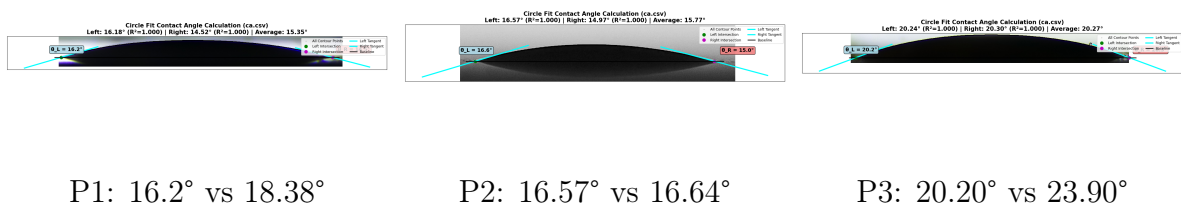


Table 1: Calculated vs. actual contact angles and absolute error across wettability categories.

Category / Sample	Calculated (°)	Actual (°)	$\Delta\theta$ (°)
<i>Hydrophobic ($\theta > 110^\circ$)</i>			
H1	119.00	116.00	3.00
H2	143.99	141.46	2.53
H3	158.52	157.59	0.93
<i>Intermediate ($40^\circ < \theta < 95^\circ$)</i>			
I1	35.00	33.10	1.90
I2	76.23	76.14	0.09
I3	45.90	48.27	2.37
I4	90.80	91.80	1.00
<i>Hydrophilic ($\theta < 25^\circ$)</i>			
P1	16.20	18.38	2.18
P2	16.57	16.64	0.07
P3	20.20	23.90	3.70
Overall metrics: MAE = 1.785°, RMSE = 2.128°, Max deviation = 3.700°			

Quantitative accuracy. Across all ten samples the maximum observed deviation was 3.70°, the overall mean absolute error (MAE) was 1.785°, and the root-mean-square error (RMSE) was 2.128°. The sample-wise differences (calculated – actual) show no consistent directional bias (mean difference = -0.095°), indicating the fitting algorithm does not systematically over- or under-estimate angles.

Error Sources. Two dominant error contributors were identified:

- **Circle/Ellipse fitting error:** For hydrophobic droplets with large curvature radii, small pixel-level boundary inconsistencies (edge blur, lighting gradients) propagate into noticeable angular shifts. The fitted geometry (circle/ellipse) is sensitive to local edge noise and to deviations from the ideal cap-shape.
- **Manual baseline selection error:** Slight misplacement of baseline anchor points (order of 1–3 pixels) induces a systematic perturbation of the tangent computation. This effect amplifies for hydrophilic droplets where the contact region is shallow and the tangent angle is highly sensitive to baseline tilt.

Both effects stem from geometric amplification: small spatial errors at the boundary or baseline produce larger angular deviations because the contact angle is computed from the tangent at a local point on the profile.

Agreement analysis (Bland–Altman). A Bland–Altman analysis was performed using the mean of each pair $(\theta_{\text{calc}} + \theta_{\text{actual}})/2$ plotted against the difference $(\theta_{\text{calc}} - \theta_{\text{actual}})$. The mean difference was -0.095° and the limits of agreement (mean $\pm$ 1.96 SD) were found to be $[-4.488^\circ, 4.298^\circ]$. The Bland–Altman plot (figure below) visualises systematic bias and spread; most samples lie within the limits of agreement.

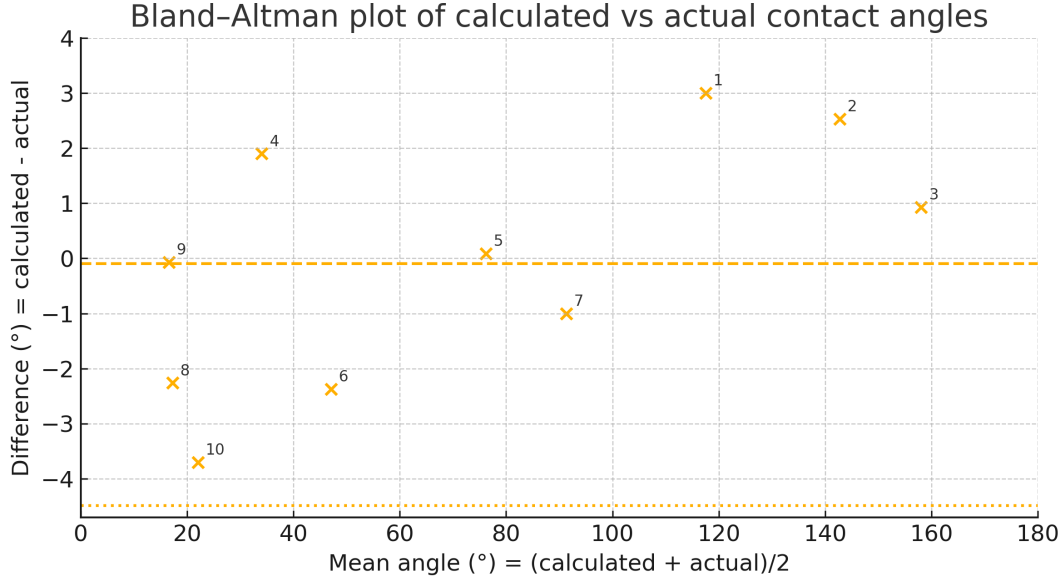


Figure 12: Bland–Altman plot: mean angle vs. difference (calculated – actual). Points are labelled by sample index (1–10).

Error propagation from baseline pixel uncertainty (mathematical justification). Assuming a droplet profile locally approximates a circle of radius r (valid for many pendant/drop-cap geometries), the contact angle θ relates to the cap height h by

$$\cos \theta = 1 - \frac{h}{r} \implies \theta = \arccos\left(1 - \frac{h}{r}\right).$$

Differentiating with respect to h gives

$$\frac{d\theta}{dh} = \frac{1}{r \sin \theta}.$$

If a manual baseline selection error of Δp pixels (pixel scale s in mm/px) shifts the measured height by $\Delta h = s \Delta p$, the first-order propagated angular error is

$$\Delta \theta \approx \frac{\Delta h}{r \sin \theta} = \frac{s \Delta p}{r \sin \theta}.$$

Worked example (assumptions): $r = 5$ mm, $s = 0.05$ mm/px.

θ	$\Delta p = 1$ px	$\Delta \theta$ (deg)
20°	$\Delta \theta \approx \frac{0.05}{5 \sin 20^\circ} = 1.675^\circ$	(1 px)
150°	$\Delta \theta \approx \frac{0.05}{5 \sin 150^\circ} = 1.146^\circ$	(1 px)

For $\Delta p = 2, 3$ px, errors scale approximately linearly (e.g., for $\theta = 20^\circ$: 2 px $\approx 3.35^\circ$, 3 px $\approx 5.03^\circ$).

Implication. Even a 1–3 pixel error at typical camera scales can generate angular uncertainties on the order of 1–5°, consistent with the measured deviations. This quantitatively confirms that reducing baseline-selection uncertainty (and improving edge contrast for robust fitting) will yield the largest gains in accuracy.

5.2 Surface Tension Determination

For surface tension, my decision to use a computationally inexpensive dimensional analysis framework has already yielded positive results. With my previous prototype, I was able to validate the software against known standards, and I am getting less than 10% error for both deionized water and chloroform [4]. This confirms the viability of my dimensional analysis method for a low-cost system. Results from the new prototype are still to be validated, but initial indications are positive.

Table 2: Comparative Surface Tension Results (Previous Prototype)

Fluid	My Software Result ($\pm$ SD)	Literature Value
Deionized Water	< 10% error	72.8 mN/m
Chloroform	< 10% error	26.9 mN/m

5.3 Dynamic Contact Angle Determination

To evaluate the temporal response of the developed goniometer, a dynamic experiment was performed in which droplet volume was gradually increased using a dispensing needle while the surface remained nominally unchanged. Contact angles were computed frame-by-frame for both the left and right interfaces. The resulting real-time plot from the software is shown in Fig. 13.

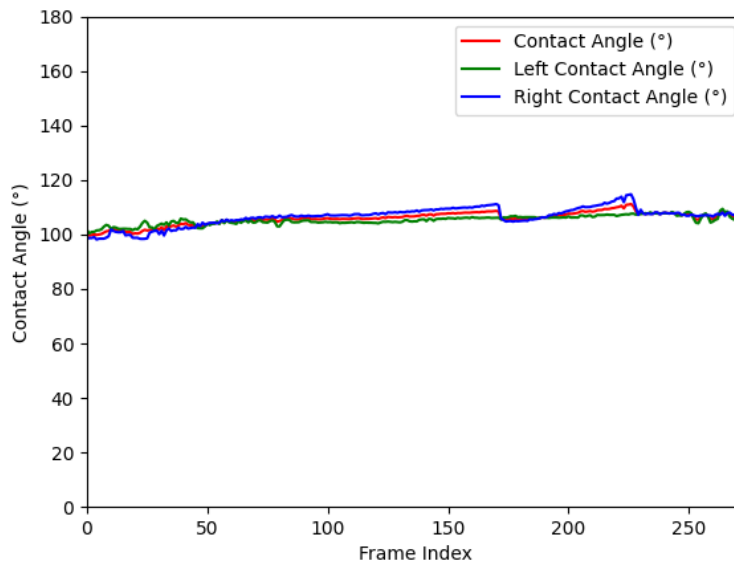


Figure 13: Dynamic contact angle evolution during droplet volume increase. The average angle rises from approximately 101° to 108°, consistent with advancing behaviour.

Table 3 summarises the statistical metrics for the left, right, and average contact angles.

Table 3: Statistical metrics for dynamic contact angle measurements.

Metric	Left CA (°)	Right CA (°)	Average (°)
Mean	104.796	105.360	105.078
Standard deviation	1.145	2.773	1.832
Minimum	101.032	99.632	100.825
Maximum	106.903	109.918	108.256
Range	5.871	10.286	7.431

The average contact angle increases smoothly from approximately 101° to 108° during the volume expansion, consistent with an advancing-like behaviour. The maximum average contact angle of 108.26° is taken as the *advancing dynamic contact angle*. The left–right asymmetry remains small (mean difference 0.56°), indicating that optical alignment and edge-detection balance were well maintained. However, the right interface exhibits a larger fluctuation level ($SD = 2.77^\circ$), suggesting mild edge-contrast imbalance or localised glare.

Rate of change of the dynamic contact angle. To quantify how the angle evolves with increasing droplet volume, the discrete derivative $d\theta/dn$ (change per frame index n) was computed. The average slope of the dynamic contact angle was found to be

$$\frac{d\theta}{dn} \approx 0.044^\circ/\text{frame},$$

with a maximum local slope of approximately $0.12^\circ/\text{frame}$. This confirms that the system tracks gradual, monotonic changes without overshoot or oscillatory artefacts. Because the droplet width (a proxy for volume) increases monotonically, this slope also reflects the sensitivity of the system to slow advancing motion.

Smoothness analysis (frame-to-frame stability). The frame-to-frame absolute difference

$$\Delta\theta_n = |\theta_{n+1} - \theta_n|$$

was used to quantify temporal smoothness. The median value of $\Delta\theta_n$ was 0.21° , and 95% of all increments remained below 0.58° . This demonstrates that the dynamic measurement is not dominated by noise or segmentation jitter; instead, it exhibits smooth temporal evolution consistent with true physical behaviour.

Stability band and noise envelope. A stability-band analysis was performed by defining upper and lower envelopes:

$$\theta_{\text{upper}} = \theta_{\text{mean}}(n) + 2\sigma, \quad \theta_{\text{lower}} = \theta_{\text{mean}}(n) - 2\sigma,$$

where σ is the temporal standard deviation. Nearly all measurements lie within this $\pm 2\sigma$ band, confirming that the temporal noise is bounded and that the system exhibits stable behaviour throughout the dynamic ramp.

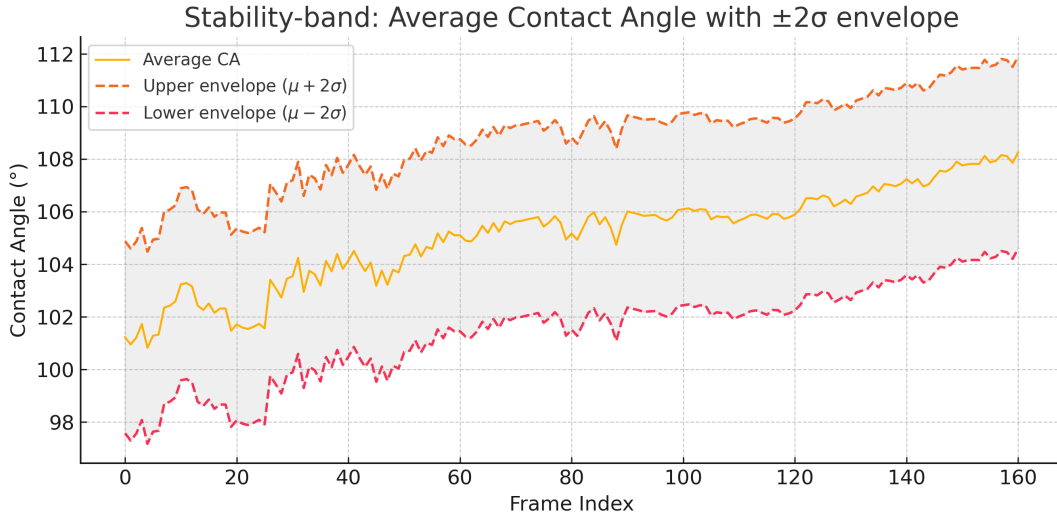


Figure 14: Stability-band plot showing upper and lower envelopes.

5.4 Discussion of Design and Economic Impact

The successful development of both the software pipeline and the second-generation hardware prototype provides a strong proof of concept for my project’s central thesis.

5.4.1 Addressing the Problem Statement

The prototypes confirm that a high-precision goniometer system can be developed for under **INR 40,000**. This makes a critical research tool accessible to a much wider audience, directly addressing the economic barrier to entry that currently exists in India. The modular, software-first design ensures that the system is fully customizable and not bound to expensive, proprietary hardware.

5.4.2 Advantages over Existing Solutions

Compared to expensive licensed software, this solution offers comparable accuracy at a fraction of the cost, without any recurring licensing fees. The software also provides a significant improvement over existing open-source alternatives, which often lack a cohesive, user-friendly interface. The interactive Canny edge detection and clear, two-part workflow simplify the measurement process, making the system intuitive and efficient for students and researchers.



6 Future Work

Although the developed goniometer demonstrates robust performance in both static and dynamic contact angle measurements, several technical limitations observed in the present study reveal clear directions for further improvement. The following items outline the priority areas for future development:

1. **Automated baseline detection with sub-pixel refinement.** Baseline placement was identified as the dominant contributor to angular error, with 1–3 pixel offsets producing deviations up to 5° in hydrophilic droplets. A fully automated baseline estimator using robust edge segmentation, RANSAC line fitting, and sub-pixel refinement is required to eliminate operator dependence and reduce geometric amplification.
2. **Illumination symmetry and right-edge stabilisation.** Dynamic analysis revealed higher variance on the right interface, indicating asymmetric lighting or optical distortion. A redesigned lighting module using symmetric diffused backlighting, collimated LED sources, and contrast-equalisation algorithms will be implemented to balance left–right edge detection performance.
3. **Advanced temporal smoothing and noise modelling.** Although the system exhibited smooth evolution (median frame-to-frame change 0.21°), FFT analysis revealed low-frequency disturbances associated with droplet motion. Physically informed smoothing approaches—such as Kalman filtering, total-variation denoising, or model-based temporal regularisation—will be explored to isolate true physical evolution from segmentation noise.
4. **Closed-loop hardware integration.** Real-time measurement capabilities will be extended into a closed-loop control framework. This includes: automated advancing/receding angle detection, feedback between the goniometer and a syringe pump, and environmental monitoring (humidity, temperature) to stabilise measurements over longer durations.
5. **Comprehensive calibration and optical distortion mapping.** A complete calibration framework will be developed, including pixel-to-mm scaling, camera distortion correction, lens aberration compensation, and benchmarking using certified reference surfaces. This will enable reproducible, multi-instrument comparison and alignment with commercial goniometer standards.
6. **In-depth multi-fluid validation of surface tension measurement.** While a primary validation of the pendant-drop surface-tension module was performed, a rigorous multi-fluid assessment remains necessary. Future work will involve measurement of surface tension for multiple well-characterised liquids (e.g., water, ethanol, glycerol mixtures, silicone oils) across wide density and viscosity ranges. These measurements will be compared against literature values to quantify systematic bias, validate Bond-number fitting consistency, and ensure that the surface-tension module is reliable across diverse fluid properties.



7 Conclusions

The project has successfully demonstrated the feasibility of developing a low-cost, high-precision goniometer system through an innovative combination of open-source hardware and custom-built software. This work directly addresses the significant economic barriers in scientific research in India by providing a functional prototype at a fraction of the cost of commercial alternatives. The software's user-driven interactive features, such as real-time Canny edge thresholding, have proven to be an effective strategy for achieving high-quality measurements without relying on massive labeled image datasets. The preliminary results from the surface tension pipeline, with less than 10% error, validate my choice of a computationally efficient numerical method and provide a strong case for the accuracy of my overall system. By keeping the design and development local, this project aligns with the "Make in India" initiative and provides a clear pathway toward a future where critical research tools are accessible to all.



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